

References

SWE101 — Topic 2: New Skills Required in the AI Era

All sources used in `research.md` and `reflection.md` are listed below.

1. Gartner — Generative AI Will Require 80% of Engineering Workforce to Upskill Through 2027

- **Publisher:** Gartner, Inc.
 - **Date:** October 3, 2024
 - **URL:** <https://www.gartner.com/en/newsroom/press-releases/2024-10-03-gartner-says-generative-ai-will-require-80-percent-of-engineering-workforce-to-upskill-through-2027>
 - **Key data used:** 80% of engineering workforce must upskill by 2027; 56% of engineering leaders cite AI/ML as biggest skills gap; prompt engineering and RAG identified as essential skills.
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2. Stack Overflow — 2025 Developer Survey (AI Section)

- **Publisher:** Stack Overflow
 - **Date:** 2025
 - **URL:** <https://survey.stackoverflow.co/2025/ai/>
 - **Key data used:** 46% of developers distrust AI accuracy vs 33% who trust it; 76% of developers resist AI for deployment/monitoring; 69% resist AI for project planning.
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3. Stack Overflow — 2025 Developer Survey Results (Blog)

- **Publisher:** Stack Overflow Blog

- **Date:** December 29, 2025
 - **URL:** <https://stackoverflow.blog/2025/12/29/developers-remain-willing-but-reluctant-to-use-ai-the-2025-developer-survey-results-are-here/>
 - **Key data used:** 69% of developers learned new skills in the past year; 44% used AI-assisted tools to do so, up from 37% in 2024.
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4. PwC — 2025 Global AI Jobs Barometer

- **Publisher:** PricewaterhouseCoopers (PwC)
 - **Date:** 2025
 - **Referenced via:** <https://k2university.com/news/why-prompt-engineering-is-now-a-core-it-skill/>
 - **Key data used:** Workers with AI skills including prompt engineering earn a 56% wage premium over peers without those skills, doubling from 25% the prior year.
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5. K2 University — Why Prompt Engineering is Now a Core IT Skill

- **Publisher:** K2 University
 - **Date:** 2025–2026
 - **URL:** <https://k2university.com/news/why-prompt-engineering-is-now-a-core-it-skill/>
 - **Key data used:** Technical architect demand grew 27% against only 4% supply growth; AI is absorbing routine execution work and moving value to the design layer.
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6. IBM Think — The 2026 Guide to Prompt Engineering

- **Publisher:** IBM
 - **Date:** 2026
 - **URL:** <https://www.ibm.com/think/prompt-engineering>
 - **Key data used:** Prompt engineering is described as a critical and growing skill; AI is now a system component in enterprise platforms, not just an external tool.
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7. Thoughtworks — Software Engineering Skills, Jobs and Careers in the AI Era

- **Publisher:** Thoughtworks
 - **Date:** 2025
 - **URL:** <https://www.thoughtworks.com/insights/articles/software-engineering-skills-jobs-careers-ai-era>
 - **Key data used:** Future developers described as “stewards of systems”; concerns raised about junior developers losing foundational knowledge by over-relying on AI tools.
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8. IEEE Spectrum — AI Jobs in 2025: Essential Insights for Software Engineers

- **Publisher:** IEEE Spectrum (Stanford HAI 2025 AI Index Report)
 - **Date:** April 17, 2025
 - **URL:** <https://spectrum.ieee.org/ai-jobs-in-2025>
 - **Key data used:** U.S. AI job postings requiring AI skills rose to 1.8% of all postings in 2024, up from 1.4% in 2023; Python remains the top AI programming skill.
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9. Arxiv — Prompt Engineer: Analyzing Skill Requirements in the AI Job Market

- **Publisher:** Academic preprint (arxiv.org)
 - **Date:** May 29, 2025
 - **URL:** <https://arxiv.org/html/2506.00058v1>
 - **Key data used:** Communication (21.9%) and creative problem-solving (15.8%) identified as the top soft skills for prompt engineering roles.
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10. Faros AI — 2025 Report on AI-Coauthored Code Quality

- **Publisher:** Faros AI
- **Date:** 2025

- **Referenced via:** <https://sureprompts.com/blog/ai-prompts-for-engineers>
 - **Key data used:** AI-coauthored code contains approximately 1.7 times more issues per pull request than human-only code.
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Minimum 5 references required — 10 references provided.